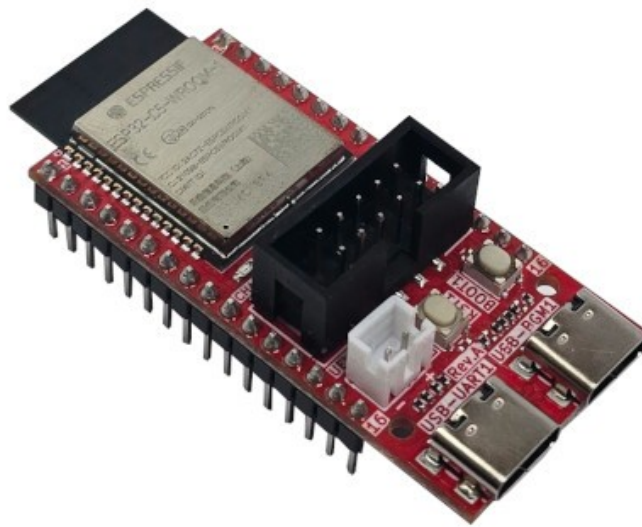




ESP32-C5-DevKit-Lipo

User Manual

olimex.com

Rev.1.0 November 2025

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What is ESP32-C5-DevKit-Lipo

[ESP32-C5-DevKit-Lipo](#) is Dual band 2.4Ghz and 5Ghz WiFi 6, Bluetooth 5 LE, Zigbee, Thread, Matter breadboard friendly development board UEXT connector, DIL GPIO ports and LiPo battery UPS. The board is with ESP32-C5-WROOM-N8R4 module with 8MB of Flash and 4MB of RAM

The features of [ESP32-C5-DevKit-Lipo](#) are:

- ESP32-C5-WROOM-N8R4 module
- 32 bit RISC-V processor 240Mhz
- Dual-Band Wi-Fi (2.4 GHz + 5 GHz)
- Bluetooth 5, Zigbee, Thread, Matter
- 8 MB Flash + 4 MB RAM
- UEXT connector
- USB-C (Power & Debug)
- USB-C JTAG
- Boot & Reset buttons
- Extension connectors spaced at 0.9"
- LiPo UPS charger & step-up converter
- Two mounting holes
- Dimensions 53x25mm

[ESP32-C5-DevKit-Lipo](#) is Open Source Hardware, all CAD files and firmware are available, so people can study and modify.



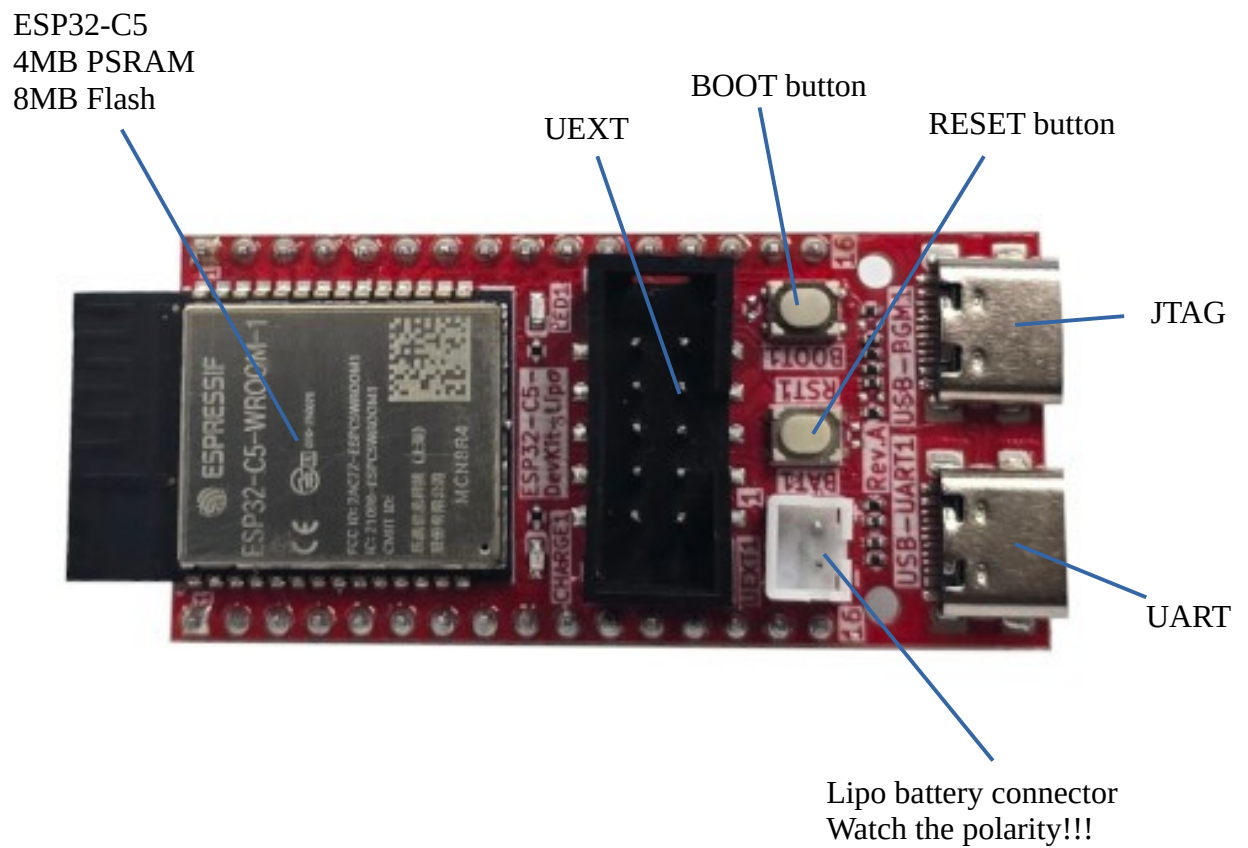
Important notice: If [ESP32-C5-DevKit-Lipo](#) is not mounted in box be careful to not place it on metal surface, nor drop metal objects on top of the PCB! This will lead to damage.

Order codes for ESP32-C5-DevKit-Lipo and accessories:

<u>ESP32-C5-DevKit-Lipo</u>	ESP32-C5 development board
<u>USB-CABLE-AM-USB3-C</u>	High speed, High current cable for power supply and programming
<u>UEXT modules</u>	many UEXT modules which can connect to UEXT connector
<u>BATTERY-LiPo1400mAh</u>	LiPo battery compatible with <u>ESP32-C5-EVB</u>
<u>BATTERY-CABLE</u>	JST2.0 cable can be used for external power supply +5V on PWR1 connector instead to power via USB-C

HARDWARE

ESP32-C5-DevKit-Lipo layout:



Power supply:

[ESP32-C5-DevKit-Lipo](#) can be powered from several sources:

USB-UART1 – this is USB-C connector with USB to serial converter attached. It works as debug UART and can be used for programming with esptool.py with automatic bootloader mode invoke.

USB-PGM1 – this is USB-C connector connected directly to ESP32-C5 JTAG ports.

J1-14/15 – J1 and J2 are 0.1” step connectors exposing all available GPIOs. The 5V power is present there on pin 14 is +5V on pin 15 is GND



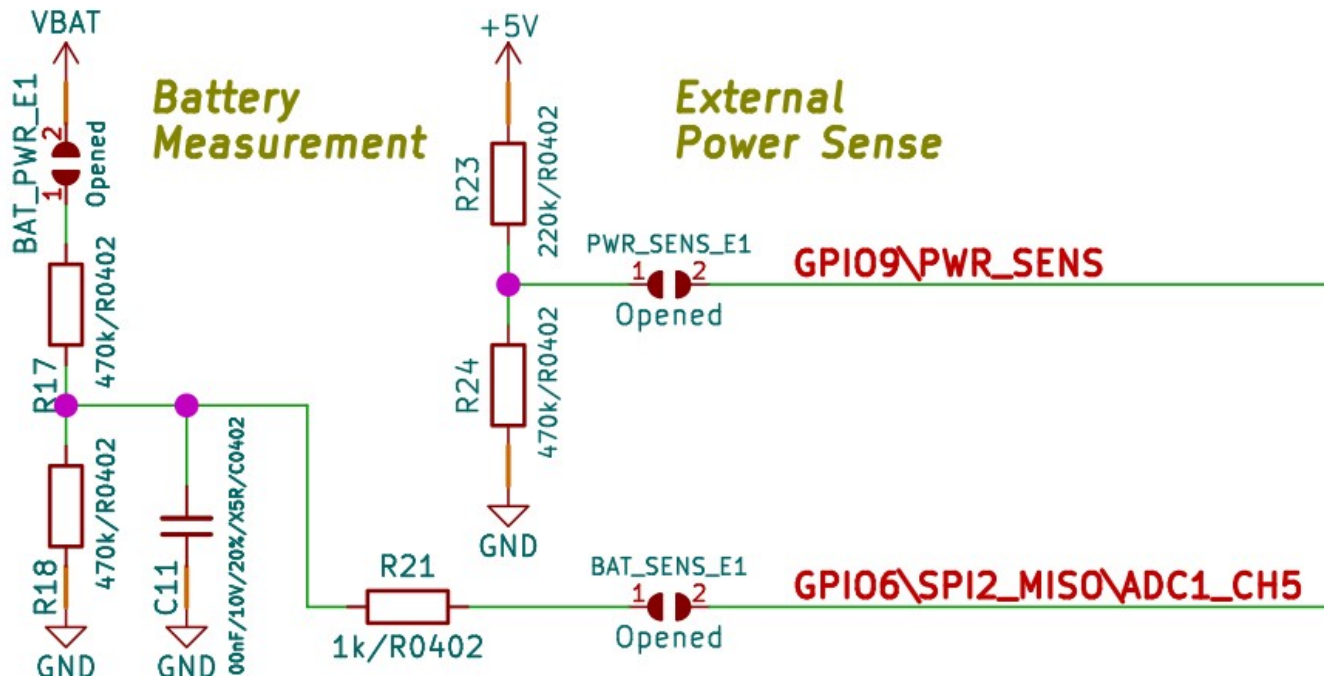
Important notice: Never connect more than 5V on J1-14/15! This will damage the board



Important notice: The USB-C VCC is directly connected to J1-14 +5V so you should not apply external power supply to this pin while you are connected to USB-C connector, or there is risk to damage your USB-host.

LiPo battery – [ESP32-C5-DevKit-Lipo](#) have smart switching between the different power supplies and LiPo charger, so when USB-C and PWR1 external power supply is missing LiPo battery with Step-Up connector is used to keep all [ESP32-C5-DevKit-Lipo](#) functionality.

When working on LiPo battery there are provisions to sense if the board is powered by LiPo battery or external source with GPIO9 and the battery voltage with GPIO6. By default these are not connected so GPIO9 and GPIO6 can be used for other purpose, but SMT jumper can be shorted in case this functionality is necessary.



UEXT connector:

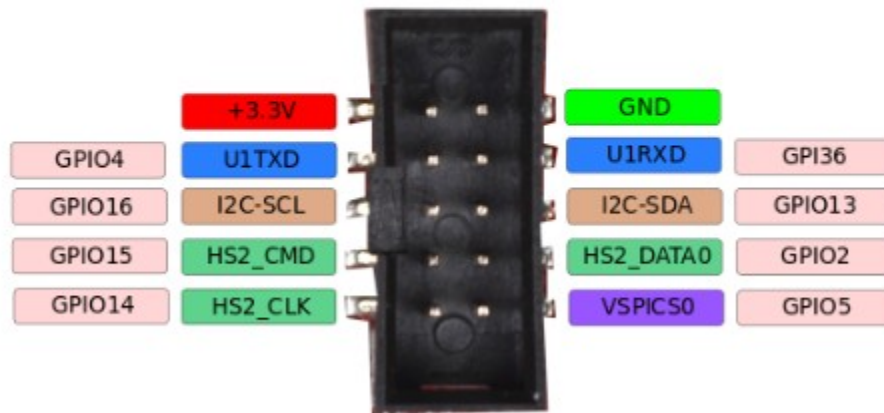
UEXT connector stands for Universal EXTension connector and contain +3.3V, GND, I2C, SPI, UART signals.

UEXT connector can be in different shapes.

The original UEXT connector is 0.1" 2.54mm step boxed plastic connector. All signals are with 3.3V levels.

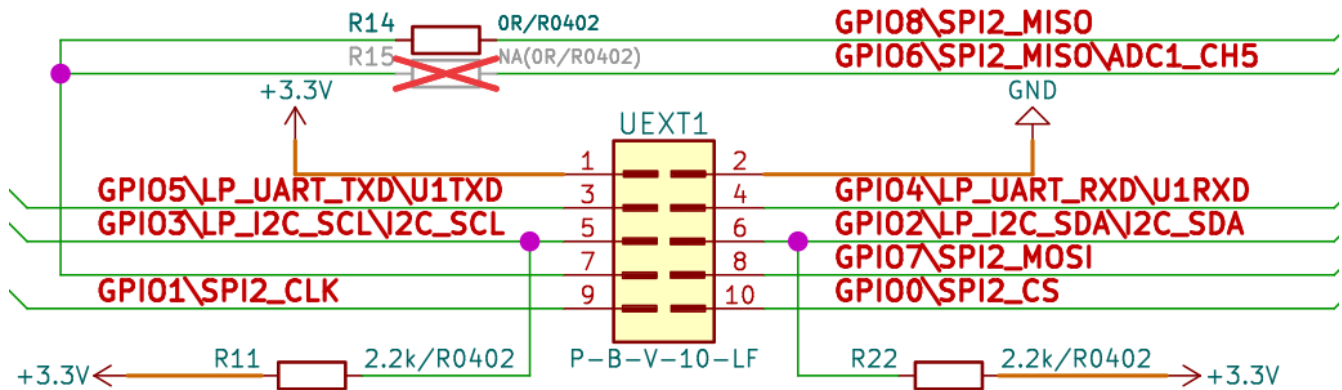
UEXT connector

note it share same pins with EXT1 and EXT2

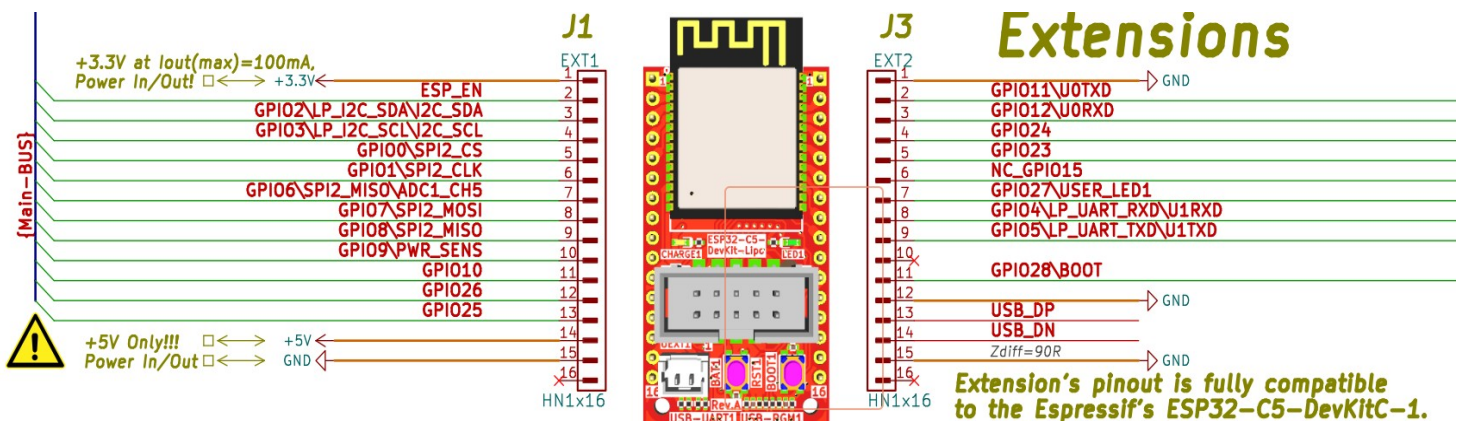


Olimex has developed number of [MODULES](#) with this connector. There are temperature, humidity, pressure, magnetic field, light sensors. Modules with LCDs, LED matrix, Relays, Bluetooth, Zigbee, WiFi, GSM, GPS, RFID, RTC, EKG, sensors and etc.

ESP32-C5-DevKit-Lipo UEXT connector:

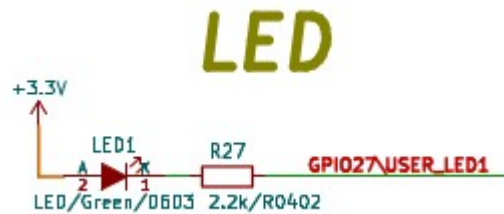


J1 and J2 connectors signals:



LEDs:

GPIO27 is connected to GREEN user LED, when the level is LOW LED is ON, when the level is HIGH the LED is OFF.



YELLOW LED for CHARGING status: OFF – battery is fully charged, ON – battery is charging, Blink – No battery.

ESP32-C5-DevKit-Lipo schematics:

[ESP32-C5-DevKit-Lipo](#) latest schematic is on [GitHub](#)

SOFTWARE:

[ESP32-C5-DevKit-Lipo](#) can be programmed with Espressif IDF 5.5 and Arduino.

Revision History

Revision 1.0 November 2025